CAI6734 Weekly Report 4: LA-CDM Training and EHR Search

Omer Kahveci

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1 Project Objective

LA-CDM reproduction. This week I continued the LA-CDM reproduction using Meta-Llama-3-8B-Instruct and the four-condition MIMIC clinical decision-making cohort. The immediate goal was to complete one 300-step training cycle despite the memory limit of the two NVIDIA L4 GPUs available through the course allocation.

OMOP EHR agent. In parallel, I continued the OMOP EHR patient-search prototype, focusing on concept vocabulary coverage, more reliable search decisions, and ways to present results.

2 Accomplishments

LA-CDM reproduction. The training schedule assigned 100 optimizer steps each to Decision Agent group-relative policy optimization (GRPO), Hypothesis Agent supervised fine-tuning, and Hypothesis Agent confidence-calibration GRPO. One L4 GPU held the trainable model and the other ran vLLM to generate clinical trajectories. The final job used four generations per case, a forward/backward microbatch size of one, a 128-token limit per generated completion, and a 2,048-token trajectory limit. The model used LoRA adapters, BF16 computation, and non-reentrant gradient checkpointing.

To fit the training GPU's memory, I changed the trainer to split each GRPO group into one-example forward and backward passes, compute reference-model log probabilities in no-gradient chunks, and calculate only the loss active in the current training phase. For Hypothesis Agent fine-tuning, I restricted the scored logits to the short supervised suffix instead of the full prompt. I also enabled non-reentrant gradient checkpointing and reduced the maximum trajectory length from 4,096 to 2,048 tokens. Checkpoints every 25 steps allowed interrupted runs to resume without repeating completed training.

The resumed run completed all 300 optimizer steps and ended with a clean Slurm finish line on October 20. The final validation pass covered 240 cases with four rollouts per case, or 960 predictions. Table 1 places these results beside the earlier untrained zero-shot test run, which used one rollout for each of 240 test cases.

The reported training loss was 0.005625. The final validation took about 2.48 hours. The large difference in all-prediction accuracy is encouraging, but the different splits and rollout counts prevent attributing that difference to training alone.

For context, the original LA-CDM paper trained Qwen-2.5-7B-Instruct and reported the test-set results in Table 2. The paper trained until validation convergence, taking approximately three days; our Llama run completed one 300-step cycle. The different model, training duration, and

Table 1: Untrained test results and trained validation results. These use different data splits and rollout counts and are descriptive, not a controlled before-and-after comparison.

Metric	Untrained test	Trained validation
Diagnosis accuracy, all predictions	8.75%	56.46%
Diagnosis accuracy, diagnosed only	72.41%	67.50%
Undiagnosed fraction	87.92%	16.35%
Action-format reward	0.1208	0.8365
Macro F1, all predictions	15.54%	57.59%
Hypothesis accuracy	65.25%	63.06%
Expected calibration error	10.87%	7.89%

evaluation setup make these published values reference points rather than direct targets for our validation run (Bani-Harouni et al., 2026, Table 1, Section 6, and Appendix C).

Table 2: Published trained Qwen LA-CDM test results.

Metric	Paper result
Mean class accuracy	81.3%
Micro F1	84.1%
Macro F1	81.3%
Average test cost	$1,295.61
Hypothesis accuracy	81.9%
Expected calibration error	3.7%

OMOP EHR agent. Dataset analysis revealed gaps in the concept names supplied with the MIMIC demo data: some IDs in the patient records could not be resolved using its local vocabulary alone. I downloaded the Athena vocabulary and incorporated its concepts and concept-ancestor relationships during preprocessing. This provides names for more IDs and lets the search consider broader measurement groups as well as individual recorded concepts.

I also experimented with a more structured search agent. It first chooses the relevant tables, time scope, and whether the request needs source records or numeric analysis. For measurement requests, it examines patient-specific candidate concepts and ancestor groups before selecting IDs. The application records these decisions so they can be inspected alongside the returned evidence.

I explored ways to visualize the results in the interface. Measurement analysis now produces values over time, distributions, or summary statistics from retrieved numeric rows, with line and bar charts for the first two views. The interface also shows the selected concepts and their row counts, which helps explain empty or partial results. These are exploratory improvements to the prototype rather than a measured gain in search accuracy.

3 Challenges

LA-CDM reproduction. Earlier jobs exceeded the training GPU's roughly 22 GB usable memory on long trajectories, including an out-of-memory failure during backpropagation at step 90. Once the memory changes allowed longer runs, the Hypothesis Agent phase exposed a separate

batching error: its target rows did not match the four GRPO rollout rows. I corrected its microbatch indexing and weighted each microbatch by its share of supervised target tokens. The completed run passed both loss-phase transitions.

Some trajectories still contained invalid action formats or unknown test names; the 16.35% undiagnosed fraction shows that completing the action protocol remains a meaningful limitation.

OMOP EHR agent. The EHR search still depends on identifying the intended concept IDs from a natural-language question. Vocabulary names, source values, and ancestor groups provide useful candidates, but similarly named concepts and concepts missing from a patient's candidate list can still lead to an incomplete or unsupported result. I have not yet measured concept-selection accuracy on a set of manually checked patient questions.

4 Next Steps

LA-CDM reproduction. This result establishes that the full three-phase pipeline can train and resume on the course GPU allocation. The side-by-side figures do not yet establish improvement against the zero-shot baseline under matched evaluation conditions. I will verify the saved final adapter and evaluate it on the fixed 240-case test set using the baseline evaluation settings. I will then compare accuracy, undiagnosed fraction, format reward, test use, and calibration, and inspect representative invalid-action examples.

OMOP EHR agent. I will improve concept-ID identification by reviewing failed and ambiguous selections, comparing vocabulary names with source values and ancestor matches, and testing candidate ranking against manually verified questions. I will use those checks to refine how candidate concepts are presented to the model and how uncertain selections are reported before expanding the search workflow further.

5 AI Disclaimer

I used OpenAI Codex to help interpret logs, revise training and EHR search code, and draft this report. I reviewed the technical claims and remain responsible for the final report.